

```

void correl(VecDoub_I &data1, VecDoub_I &data2, VecDoub_0 &ans) {
Computes the correlation of two real data sets data1[0..n-1] and data2[0..n-1] (including
any user-supplied zero padding). n must be an integer power of 2. The answer is returned in
ans[0..n-1] stored in wraparound order, i.e., correlations at increasingly negative lags are in
ans[n-1] on down to ans[n/2], while correlations at increasingly positive lags are in ans[0]
(zero lag) on up to ans[n/2-1]. Sign convention of this routine: if data1 lags data2, i.e., is
shifted to the right of it, then ans will show a peak at positive lags.
    Int no2,i,n=data1.size();
    Doub tmp;
    VecDoub temp(n);
    for (i=0;i<n;i++) {
        ans[i]=data1[i];
        temp[i]=data2[i];
    }
    realft(ans,1);
    realft(temp,1);
    no2=n>>1;
    for (i=2;i<n;i+=2) {
        tmp=ans[i];
        ans[i]=(ans[i]*temp[i]+ans[i+1]*temp[i+1])/no2;
        ans[i+1]=(ans[i+1]*temp[i]-tmp*temp[i+1])/no2;
    }
    ans[0]=ans[0]*temp[0]/no2;
    ans[1]=ans[1]*temp[1]/no2;
    realft(ans,-1);
}

```

correl.h

Transform both data vectors.

Normalization for inverse FFT.

Multiply to find FFT of their correlation.

Inverse transform gives correlation.

The *discrete autocorrelation* of a sampled function g_j is just the discrete correlation of the function with itself. Obviously this is always symmetric with respect to positive and negative lags. Feel free to use the above routine `correl` to obtain autocorrelations, simply calling it with the same data vector in both arguments. If the inefficiency bothers you, you can edit the program so that only one call is made to `realft` for the forward transform.

CITED REFERENCES AND FURTHER READING:

Brigham, E.O. 1974, *The Fast Fourier Transform* (Englewood Cliffs, NJ: Prentice-Hall), §13–2.

13.3 Optimal (Wiener) Filtering with the FFT

There are a number of other tasks in numerical processing that are routinely handled with Fourier techniques. One of these is filtering for the removal of noise from a “corrupted” signal. The particular situation we consider is this: There is some underlying, uncorrupted signal $u(t)$ that we want to measure. The measurement process is imperfect, however, and what comes out of our measurement device is a corrupted signal $c(t)$. The signal $c(t)$ may be less than perfect in either or both of two respects. First, the apparatus may not have a perfect delta-function response, so that the true signal $u(t)$ is convolved with (smeared out by) some known response function $r(t)$ to give a smeared signal $s(t)$,

$$s(t) = \int_{-\infty}^{\infty} r(t - \tau)u(\tau) d\tau \quad \text{or} \quad S(f) = R(f)U(f) \quad (13.3.1)$$

where S, R, U are the Fourier transforms of s, r, u , respectively. Second, the measured signal $c(t)$ may contain an additional component of noise $n(t)$,

$$c(t) = s(t) + n(t) \quad (13.3.2)$$

We already know how to deconvolve the effects of the response function r in the absence of any noise (§13.1); we just divide $C(f)$ by $R(f)$ to get a deconvolved signal. We now want to treat the analogous problem when noise is present. Our task is to find the *optimal filter*, $\phi(t)$ or $\Phi(f)$, which, when applied to the measured signal $c(t)$ or $C(f)$ and then deconvolved by $r(t)$ or $R(f)$, produces a signal $\tilde{u}(t)$ or $\tilde{U}(f)$ that is as close as possible to the uncorrupted signal $u(t)$ or $U(f)$. In other words, we will estimate the true signal U by

$$\tilde{U}(f) = \frac{C(f)\Phi(f)}{R(f)} \quad (13.3.3)$$

In what sense is $\tilde{U}$ to be close to U ? We ask that they be *close in the least-square sense*

$$\int_{-\infty}^{\infty} |\tilde{u}(t) - u(t)|^2 dt = \int_{-\infty}^{\infty} |\tilde{U}(f) - U(f)|^2 df \quad \text{is minimized.} \quad (13.3.4)$$

Substituting equations (13.3.3) and (13.3.2), the right-hand side of (13.3.4) becomes

$$\begin{aligned} & \int_{-\infty}^{\infty} \left| \frac{[S(f) + N(f)]\Phi(f)}{R(f)} - \frac{S(f)}{R(f)} \right|^2 df \\ &= \int_{-\infty}^{\infty} |R(f)|^{-2} \left\{ |S(f)|^2 |1 - \Phi(f)|^2 + |N(f)|^2 |\Phi(f)|^2 \right\} df \end{aligned} \quad (13.3.5)$$

The signal S and the noise N are *uncorrelated*, so their cross product, when integrated over frequency f , gave zero. (This is practically the *definition* of what we mean by noise!) Obviously (13.3.5) will be a minimum if and only if the integrand is minimized with respect to $\Phi(f)$ at every value of f . Let us search for such a solution where $\Phi(f)$ is a real function. Differentiating with respect to Φ and setting the result equal to zero gives

$$\Phi(f) = \frac{|S(f)|^2}{|S(f)|^2 + |N(f)|^2} \quad (13.3.6)$$

This is the formula for the optimal filter $\Phi(f)$.

Notice that equation (13.3.6) involves S , the smeared signal, and N , the noise. The two of these add up to be C , the measured signal. Equation (13.3.6) does not contain U , the “true” signal. This makes for an important simplification: The optimal filter can be determined independently of the determination of the deconvolution function that relates S and U .

To determine the optimal filter from equation (13.3.6) we need some way of separately estimating $|S|^2$ and $|N|^2$. There is no way to do this from the measured signal C alone without some other information, or some assumption or guess. Luckily, the extra information is often easy to obtain. For example, we can sample a long

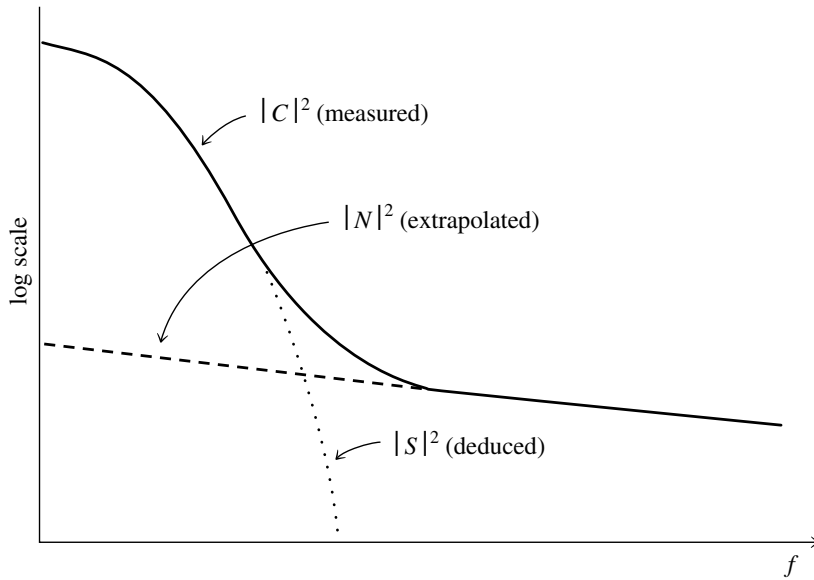


Figure 13.3.1. Optimal (Wiener) filtering. The power spectrum of signal plus noise shows a signal peak added to a noise tail. The tail is extrapolated back into the signal region as a “noise model.” Subtracting gives the “signal model.” The models need not be accurate for the method to be useful. A simple algebraic combination of the models gives the optimal filter (see text).

stretch of data $c(t)$ and plot its power spectral density using equations (12.0.15), (12.1.8), and (12.1.5). This quantity is proportional to the sum $|S|^2 + |N|^2$, so we have

$$|S(f)|^2 + |N(f)|^2 \approx P_c(f) = |C(f)|^2 \quad 0 \leq f < f_c \quad (13.3.7)$$

(More sophisticated methods of estimating the power spectral density will be discussed in §13.4 and §13.7, but the estimation above is almost always good enough for the optimal filter problem.) The resulting plot (see Figure 13.3.1) will often immediately show the spectral signature of a signal sticking up above a continuous noise spectrum. The noise spectrum may be flat, or tilted, or smoothly varying; it doesn’t matter, as long as we can guess a reasonable hypothesis as to what it is. Draw a smooth curve through the noise spectrum, extrapolating it into the region dominated by the signal as well. Now draw a smooth curve through the signal plus noise power. The difference between these two curves is your smooth “model” of the signal power. The quotient of your model of signal power to your model of signal plus noise power is the optimal filter $\Phi(f)$. [Extend it to negative values of f by the formula $\Phi(-f) = \Phi(f)$.] Notice that $\Phi(f)$ will be close to unity where the noise is negligible, and close to zero where the noise is dominant. That is how it does its job! The intermediate dependence given by equation (13.3.6) just turns out to be the optimal way of going in between these two extremes.

Because the optimal filter results from a minimization problem, the quality of the results obtained by optimal filtering differs from the true optimum by an amount that is *second order* in the precision to which the optimal filter is determined. In other words, even a fairly crudely determined optimal filter (sloppy, say, at the 10% level) can give excellent results when it is applied to data. That is why the separation